

Metric-Balanced Packet Gauges

An Exact Orthogonal Minimax Theorem, a 33-Step Obstruction, and Recovery of 132 Tail Recurrences

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Abstract

The Euclidean polar gauge used in RH-135 makes consecutive packet frames as close as possible, but only 51 of 216 recurrent nonzero-tail transitions have a contractive relative-metric recurrence in that gauge. We separate the intrinsic obstruction from the gauge artifact. For positive definite recent Gramians G, G' with increasingly ordered eigenvalues g_i, g'_i , define

$$\kappa(O) = \lambda_{\max}\left((G')^{-1/2} O^* G O (G')^{-1/2}\right), \quad O \in O(r).$$

We prove the exact orthogonal minimax formula

$$\min_{O \in O(r)} \kappa(O) = \max_i \frac{g_i}{g'_i}.$$

The same minimum is attained in both components of $O(r)$. Consequently, if the raw memory coefficient satisfies $a \max_i (g_i/g'_i) \geq 1$, then no orthogonal packet gauge can make the RH-134 Young recurrence contractive.

For forcing control we construct a deterministic 33-point family: the Euclidean polar gauge, two determinant-compatible metric-optimal eigenframe gauges, and polar projections of 17 affine interpolation points toward each endpoint. We choose the candidate minimizing the optimized affine fixed floor $q/(1-\rho)$, without claiming global optimality for that nonconvex objective. An 80-decimal audit gives an exact finite classification. Of 216 recurrent transitions, 33 are impossible for every orthogonal gauge and 183 are possible. The finite metric-balanced family recovers all 183, including 132 missed by the Euclidean polar gauge, and loses none of the original 51. All 183 optimized fixed floors are subunit; the median is 1.29×10^{-14} and the maximum is 2.514×10^{-3} . Thus the previous majority obstruction was primarily a gauge artifact, while the remaining 33 transitions form a sharp obstruction within the orthogonal moving-frame recurrence itself.

1 From Euclidean alignment to metric alignment

Let $G, D \succcurlyeq 0$ denote a recent Gramian and a memory tail in reduced packet coordinates, with $G \succ 0$. For the next time step use G', D' . RH-134 and RH-135 reduce the moving-frame

estimate to

$$D' \preceq a(1 + \tau)O^*DO + F_b + (1 + \tau^{-1})F_f(O), \quad \tau > 0, \quad (1)$$

where $a > 0$ contains the memory decay and envelope ratio, F_b is the boundary-slice forcing, and $F_f(O)$ is the frame-motion forcing. After normalization by G' , the old-tail coefficient is

$$\rho(O, \tau) = a(1 + \tau)\kappa(O), \quad \kappa(O) = \|G^{1/2}O(G')^{-1/2}\|^2.$$

The Euclidean polar gauge minimizes $\|U' - UO\|$, where U, U' are consecutive packet frames [Schonemann, 1966]. It need not minimize $\kappa(O)$ because the latter depends on anisotropic recent-Gram directions. This distinction explains the large metric factors observed in RH-135.

The quantity $\kappa(O)$ is also the least c for which

$$O^*GO \preceq cG'. \quad (2)$$

It therefore admits a complete orthogonal minimax solution using standard Loewner-order eigenvalue monotonicity [Bhatia, 1997, Horn and Johnson, 1991].

2 Exact orthogonal metric minimax

Write the increasingly ordered eigenvalues as

$$0 < g_1 \leq \dots \leq g_r, \quad 0 < g'_1 \leq \dots \leq g'_r.$$

Theorem 1 (Ordered-eigenframe minimax). *For positive definite $G, G' \in \mathbb{R}^{r \times r}$,*

$$\min_{O \in O(r)} \lambda_{\max}((G')^{-1/2}O^*GO(G')^{-1/2}) = \max_{1 \leq i \leq r} \frac{g_i}{g'_i}.$$

If $G = U \operatorname{diag}(g_i)U^$ and $G' = V \operatorname{diag}(g'_i)V^*$, then $O_* = UV^*$ attains the minimum. For either prescribed determinant sign, a minimizer exists.*

Proof. Suppose (2) holds for some c . Eigenvalue monotonicity under the Loewner order gives

$$g_i = \lambda_i(O^*GO) \leq c\lambda_i(G') = cg'_i$$

for every i , hence $c \geq \max_i g_i/g'_i$. Conversely, for $O_* = UV^*$, $O_*^*GO_* = V \operatorname{diag}(g_i)V^*$ and G' is diagonal in the same frame. Its generalized eigenvalues are exactly g_i/g'_i , proving attainment. Replacing U by US , where S is any diagonal sign matrix, does not change the aligned quadratic form. Choosing the product of the signs gives either determinant component of $O(r)$. $\square$

The proof is elementary, but it is sharp: no information about packet angles or frame smoothness can lower this value. It converts a nonconvex search over $O(r)$ into one ordered spectral ratio.

Corollary 2 (Universal orthogonal-gauge obstruction). *Let*

$$A_* = a \max_i \frac{g_i}{g'_i}.$$

If $A_ \geq 1$, then $\rho(O, \tau) > 1$ for every $O \in O(r)$ and every $\tau > 0$. If $A_* < 1$, a metric-optimal O_* and a sufficiently small $\tau > 0$ give $\rho(O_*, \tau) < 1$.*

Thus $A_* = 1$ is an exact contractivity wall for the entire RH-134 orthogonal recurrence family. It is stronger than diagnosing one failed gauge and weaker than ruling out nonorthogonal or enlarged-state recurrences.

3 Balancing metric decay against frame forcing

Metric-optimal alignment can move the frame substantially. In the present recurrence, the normalized forcing has the form

$$q(O, \tau) = q_b + B(O)(1 + \tau^{-1}), \quad B(O) = c_f \|U' - UO\|^2,$$

where q_b and c_f are fixed by the transition and target Gramian. Once $\rho < 1$, iteration gives the scalar fixed-floor objective

$$\Phi(O, \tau) = \frac{q_b + B(O)(1 + \tau^{-1})}{1 - a\kappa(O)(1 + \tau)}. \quad (3)$$

The polar gauge minimizes B , while Theorem 1 minimizes κ . Neither endpoint generally minimizes (3).

We use the following finite, reproducible compromise. Let O_p be the Euclidean polar gauge. Among the determinant-compatible sign variants of the ordered-eigenframe minimizer, select the two with smallest frame defect; call them $O_{m,1}, O_{m,2}$. For each endpoint and each $s \in \{0, 1/16, \dots, 1\}$, project $(1 - s)O_p + sO_{m,j}$ back to $O(r)$ by polar decomposition. Duplicate matrices are removed. In every audited transition this gives 33 candidates. For each candidate we optimize τ over its contractive interval and then choose the smallest finite Φ .

Proposition 3 (Contractivity completeness of the finite family). *The preceding candidate family contains a contractive gauge if and only if some orthogonal gauge is contractive.*

Proof. The family contains $O_{m,1}$ at $s = 1$. By Theorem 1 and Corollary 2, this endpoint is contractive for some τ exactly when $A_* < 1$, which is exactly the existence condition over all orthogonal gauges. $\square$

This proposition concerns feasibility only. The 33-point interpolation is not proved to find the global minimum of (3). That distinction is essential: the orthogonal minimax classification is exact, while the reported fixed floors are certified upper bounds from a finite strategy.

σ	recurrent	possible	polar	balanced
0.08	6	0	0	0
0.04	36	21	0	21
0.02	72	63	24	63
0.01	102	99	27	99
total	216	183	51	183

Table 1: Exact orthogonal feasibility versus the two finite gauge choices. The coarsest audited scale has no recurrent nonzero-tail transition.

4 Eighty-decimal packet audit

We rebuild the 30 dyadic packet chains used in RH-130–RH-135. The packet rank is four, the memory depth is eight, and the decay is $\eta = 1/512$. At each recurrent nonzero-tail transition we compute the source and target Gramians, the exact minimax ratio, both selected metric endpoints, all polar interpolants, the frame and birth forcing, and the optimized Young floor. All generalized 4×4 spectra are recomputed at 80 decimal digits, with high precision serving as a numerical audit rather than an interval proof [Higham, 2002].

There are 216 recurrent transitions. The minimax test classifies 183 as orthogonally contractive and 33 as impossible. The Euclidean polar gauge finds only 51 of the 183 possible cases. The metric-balanced family finds all 183, recovering 132 and losing none. Every recovered transition has a subunit fixed-floor bound.

The exact lower base A_* ranges from 0.002036 to 7176.13, with median 0.08996. Among feasible transitions its maximum is 0.90814, close enough to the wall that forcing control is nontrivial. The 33 universal obstructions are concentrated in all six $\sigma = 0.08$ transitions, 15 at $\sigma = 0.04$, nine at $\sigma = 0.02$, and three at $\sigma = 0.01$; 26 occur on the right channel and seven on the left.

The price of metric alignment is visible in frame motion. The median frame defect rises from 0.2215 for the polar gauge to 1.4091 for the selected balanced gauge. On feasible transitions the median defect ratio is 9.22 and the maximum is about 5.59×10^3 . Nevertheless the strong raw memory decay absorbs that cost: the 183 optimized floors range from 7.95×10^{-24} to 2.513×10^{-3} , with median 1.294×10^{-14} . Only two exceed 10^{-4} , and none reaches one. Principal-frame similarity is not a proxy for the minimax wall: its correlation with $\log_{10} A_*$ is only -0.174 on this frozen audit.

5 What is closed, and what remains open

RH-136 closes the orthogonal gauge-selection question at the level of contractive feasibility. The 165 failures reported in RH-135 split into 132 gauge artifacts and 33 genuine obstructions for every orthogonal gauge. For the 183 feasible transitions, forcing is no longer the finite bottleneck: the explicit finite candidate family gives small fixed floors throughout.

The next question is composition. A transitionwise subunit floor does not by itself give a scale-uniform normalized volume or capacity lower bound. The coefficients vary in time,

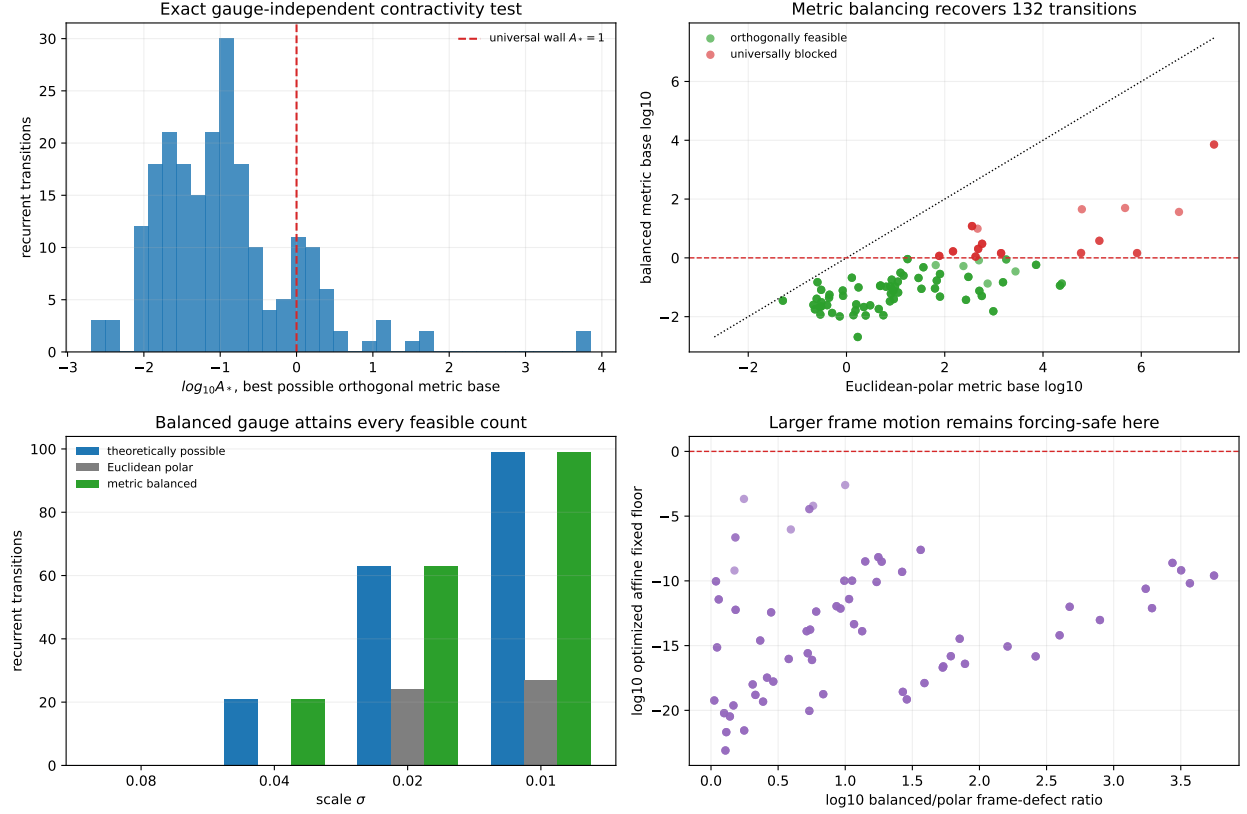


Figure 1: The minimax wall gives an exact feasibility classification. The balanced gauge reaches every feasible transition, while the forcing floor remains subunit despite larger frame displacement.

and the 33 blocked transitions must be isolated, bypassed by a reset/birth mechanism, or handled by a larger state space. A useful next invariant should therefore couple the affine tail law to recent-Gram volume and identify exactly when a blocked step destroys a multi-step certificate.

We have not proved that the 33-point family globally minimizes the affine objective, that a nonorthogonal gauge cannot bypass the 33 walls, a uniform cross-scale recurrence, a positive normalized-capacity liminf, uniform Stage A, a Hilbert–Polya operator, zeta-zero identification, or the Riemann Hypothesis.

References

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